

While GPUs were a vast improvement over CPUs, two more iterations of technology occurred to produce a more efficient chip for faster guessing of golden hashes. First came field-programmable gate arrays (FPGA), an interim chip, before the granddaddy of them all appeared: application-specific integrated circuits (ASICs). As the name implies, ASICs are application-specific, meaning that the physical hardware must be designed and manufactured with the application in mind. CPUs, GPUs, and FPGAs can all be bought generically and, with proper engineering, be applied to a specific purpose after the purchase. The physical layout of ASICs, on the other hand, needs to be etched into the chip at the semiconductor fabrication factory.

Designing and manufacturing such a specific chip requires a significant initial investment, and it was only when Bitcoin's network became big enough and bitcoin worth enough that a company could fully pursue this opportunity. The first computer—or mining rig—with ASIC chips that were specifically manufactured for the process was connected in January 2013.<sup>6</sup> Currently, top-of-the-line ASICs have a hash rate of 14 TH/s, meaning these machines crunch data and output a hash 14 trillion times a second.<sup>7</sup>

Collectively, the more computers attached to the Bitcoin network, the higher the odds of one of them discovering a golden hash. Without any adjustment, more computers would increase the supply rate of new bitcoin, leading to runaway supply inflation. For that reason, Satoshi built into Bitcoin's software the rule that as more compute power is added to the